```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
```

```
df = pd.read_csv("marketing_and_sales_data_evaluate_lr.csv")

df.head()
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

```
print(df.shape)

print(df.info())

print(df.isnull().sum())
```

```
(4572, 4)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   TV            4562 non-null   float64
 1   Radio         4568 non-null   float64
 2   Social_Media  4566 non-null   float64
 3   Sales         4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
None
TV              10
Radio            4
Social_Media     6
Sales            6
dtype: int64
```

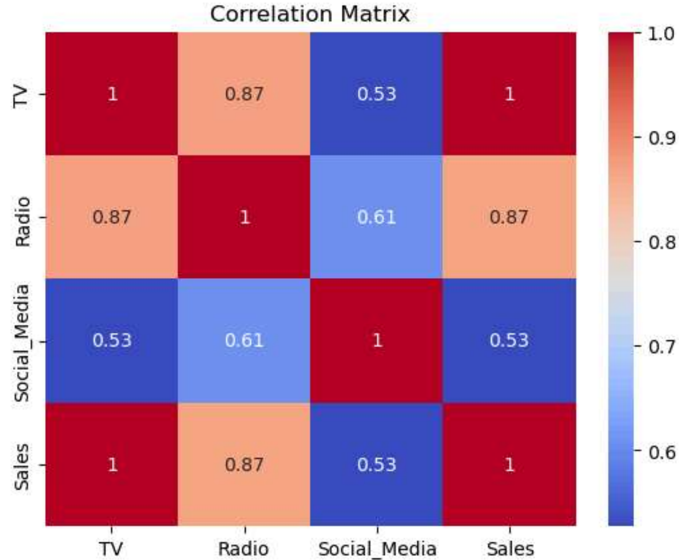
```
df = df.dropna()
```

```
print(df.isnull().sum())
```

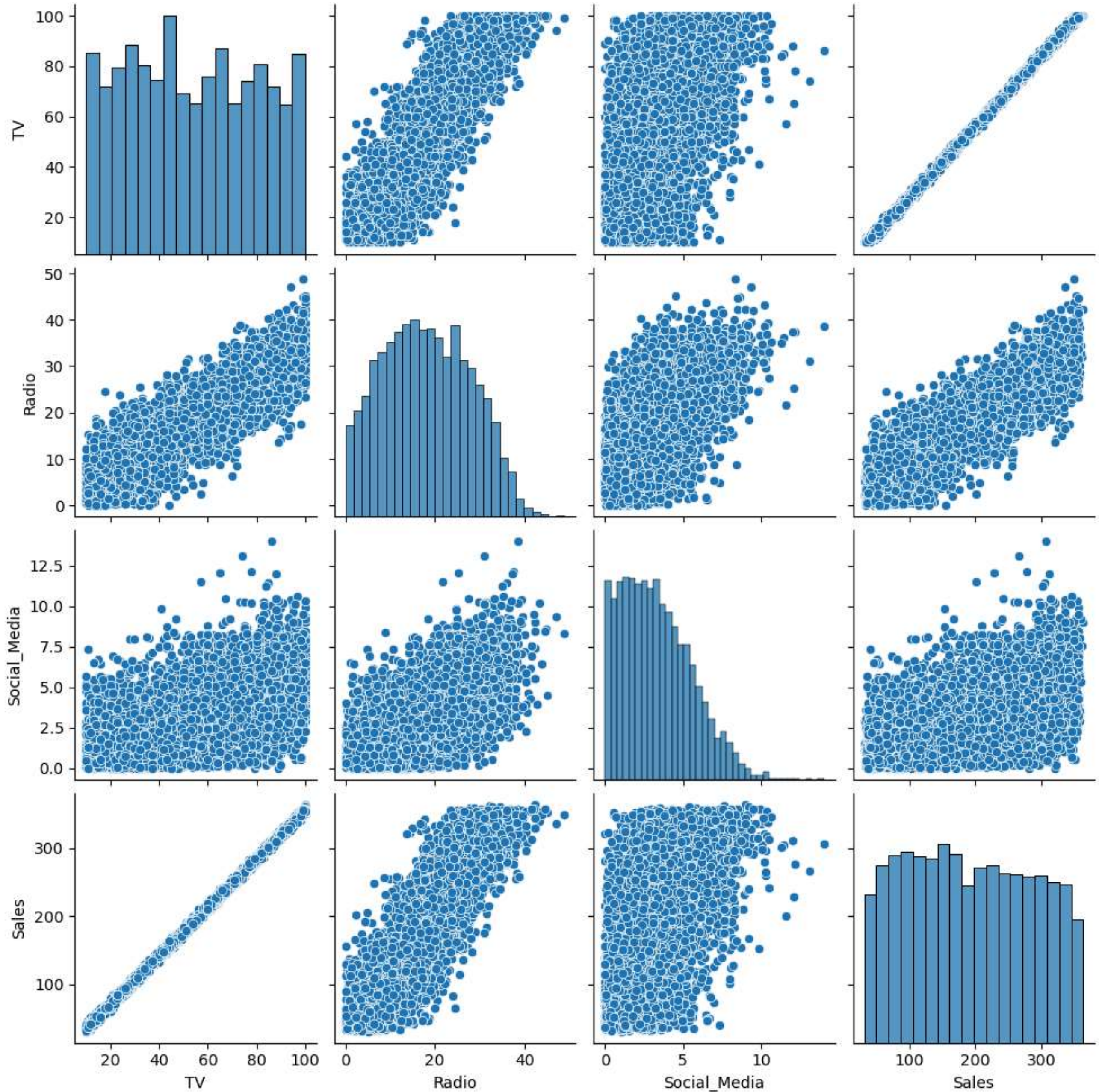
```
TV              0
Radio           0
Social_Media    0
Sales           0
dtype: int64
```

```
correlation_matrix = df.corr(numeric_only=True)

print(correlation_matrix)

sns.heatmap(correlation_matrix,
            annot=True,
            cmap="coolwarm")

plt.title("Correlation Matrix")
plt.show()
```

	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000




```
sns.pairplot(df)

plt.show()
```



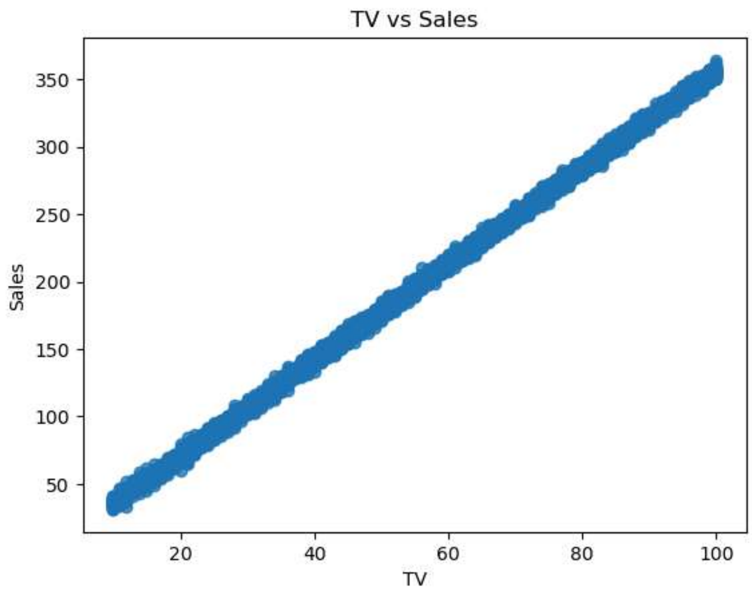

```
X = df["TV"]

y = df["Sales"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.999
Model:                            OLS   Adj. R-squared:                  0.999
Method:                 Least Squares   F-statistic:                 4.517e+06
Date:                Thu, 11 Jun 2026   Prob (F-statistic):               0.00
Time:                        07:17:31   Log-Likelihood:                -11366.
No. Observations:                4546   AIC:                         2.274e+04
Df Residuals:                    4544   BIC:                         2.275e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
```

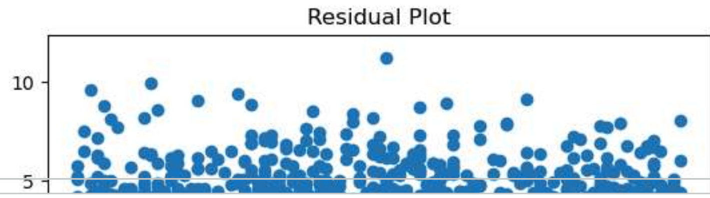
```
const         -0.1325      0.101     -1.317      0.188      -0.330       0.065
TV             3.5615      0.002   2125.272      0.000       3.558       3.565
==============================================================================
Omnibus:                        0.052   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.974   Jarque-Bera (JB):                0.031
Skew:                          -0.001   Prob(JB):                        0.985
Kurtosis:                       3.012   Cond. No.                         138.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

```
predictions = model.predict(X)

residuals = y - predictions
```

```
sns.regplot(
    x=df["TV"],
    y=df["Sales"]
)

plt.title("TV vs Sales")
plt.show()
```




```
plt.scatter(predictions, residuals)

plt.axhline(y=0)

plt.xlabel("Predicted Sales")
plt.ylabel("Residuals")

plt.title("Residual Plot")

plt.show()
```




```
sm.qqplot(residuals, line="45")

plt.show()
```